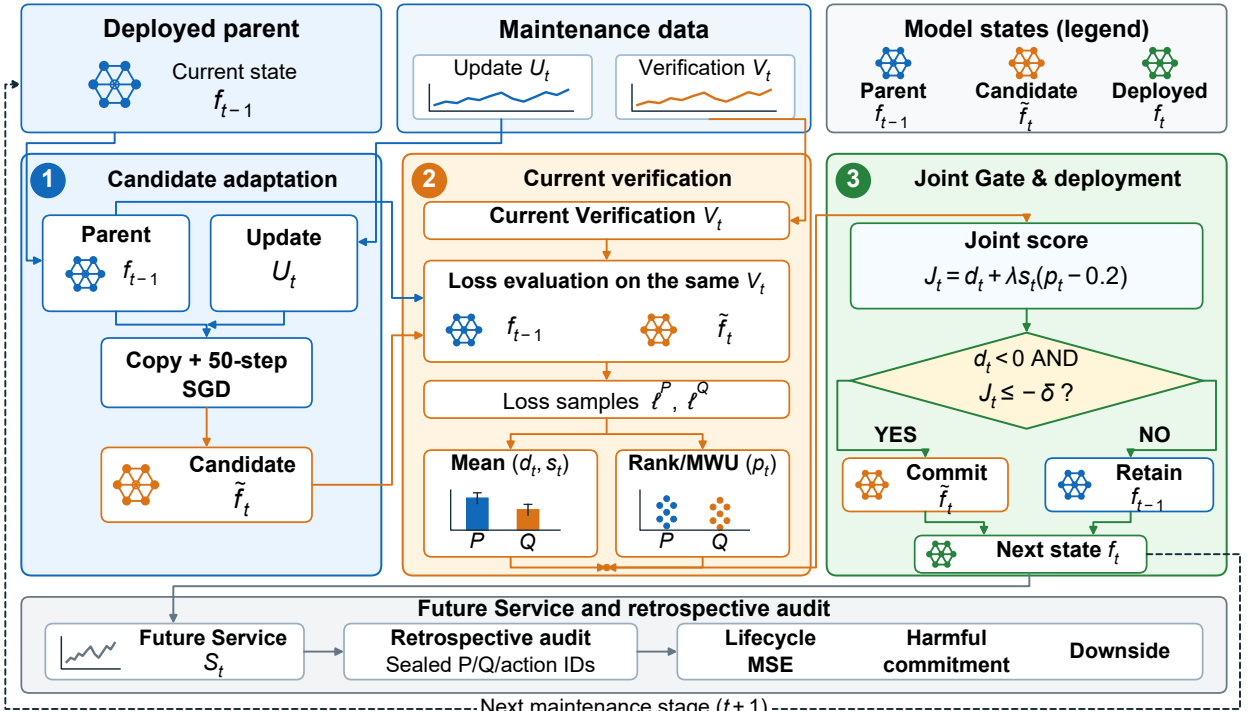

Deployed parent
Current state
f_{t-1}
Maintenance data
Update U_t
Verification V_t
Model states (legend)
Parent f_{t-1}
Candidate $\tilde{f}_t$
Deployed f_t
1
Candidate adaptation
Parent f_{t-1}
Update U_t
Copy + 50-step SGD
Candidate $\tilde{f}_t$
2
Current verification
Current Verification V_t
Loss evaluation on the same V_t
f_{t-1}
$\tilde{f}_t$
Loss samples ℓ^P, ℓ^Q
Mean (d_t, s_t)
Rank/MWU (p_t)
P
Q
3
Joint Gate & deployment
Joint score
$J_t = d_t + \lambda s_t(p_t - 0.2)$
$d_t < 0$ AND $J_t \le -\delta$?
YES
NO
Commit $\tilde{f}_t$
Retain f_{t-1}
Next state f_t
Future Service and retrospective audit
Future Service S_t
Retrospective audit
Sealed P/Q/action IDs
Lifecycle MSE
Harmful commitment
Downside
Next maintenance stage ($t+1$)